Arista Access Node - MG-ACC-38 Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) SOP

Node: MG-ACC-38 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_3330

Description

This SOP outlines the procedures for resolving critical hardware/environmental faults on Arista Access Node MG-ACC-38, Site Location: MG Road Core, Role: Access. The failure domain includes ASIC memory parity errors, line card crashes, and power supply redundancy loss, which are critical failures that must be addressed promptly to ensure network stability and availability. The underlying protocol and hardware mechanics of this failure domain are rooted in the ASIC's memory parity checking mechanism, which detects and corrects errors that occur during memory access. However, when errors persist or accumulate, the ASIC may crash, leading to line card failures or power supply redundancy loss. The primary goal of this SOP is to identify and isolate the root cause of these failures, perform necessary repairs or replacements, and verify that the system returns to a stable and operational state.

Pre-requisite Checks

- check: Verify that the MG Road Core site is accessible for remote hands and that all necessary equipment and personnel are on site.

- check: Check the redundant link status to ensure that it is operational and can take over in case of a primary link failure.

- check: Verify that the power supply redundancy is intact and functioning correctly.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes due to ASIC failure	Medium
Power supply redundancy loss due to hardware failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal error-log, Expected Output: System internal error log showing ASIC memory parity errors and line card crashes, Description: This command displays the system internal error log, which shows ASIC memory parity errors and line card crashes. This helps identify the root cause of the failure.
```

```
Command: show interfaces, Expected Output: Interface status showing line card crashes or power supply redundancy loss, Description: This command displays the interface status, which shows line card crashes or power supply redundancy loss. This helps identify the affected interface and the cause of the failure.

Command: show power supply, Expected Output: Power supply status showing redundancy loss, Description: This command displays the power supply status, which shows redundancy loss. This helps identify the cause of the failure.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: replace system internal asic , Description: Replace the faulty ASIC or line card with a new one. This requires dispatching remote hands to the MG Road Core site.

Step 2: Step: Replace the power supply, Command: replace power supply , Description: Replace the faulty power supply with a new one. This requires dispatching remote hands to the MG Road Core site.

Step 3: Step: Verify the system configuration, Command: show running-config, Description: Verify that the system configuration is correct and that there are no configuration inconsistencies that may have contributed to the failure.

Post-Check Verification

```
Command: show system internal error-log, Expected Output: System internal error log showing no ASIC memory parity errors or line card crashes, Description: Verify that the system internal error log shows no ASIC memory parity errors or line card crashes.

Command: show interfaces, Expected Output: Interface status showing no line card crashes or power supply redundancy loss, Description: Verify that the interface status shows no line card crashes or power supply redundancy loss.

Command: show power supply, Expected Output: Power supply status showing redundancy intact, Description: Verify that the power supply status shows redundancy intact.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy running-config startup-config, Description: Restore the original configuration by copying the running configuration to the startup configuration.

- Step: Replace the faulty component with the original component, Command: replace system internal asic , Description: Replace the faulty component with the original component.